Machine Learning Approach to Predict Antibiotic-Compound Synergy Against Multi-drug Resistant Bacterial Strains

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Abstract

Antibiotic resistance represents a growing global health crisis. While the development of novel antibiotics has declined, combination therapies pairing existing antibiotics with other approved drugs offer a promising strategy to combat resistance. This study presents a machine learning approach to predict antibiotic-compound synergy across multiple resistant bacterial strains. Utilizing data from a high-throughput droplet-based screen covering 1.3 million unique strain-antibiotic-compound combinations, we developed classifier models predicting whether a combination of antibiotic and compound would have a synergistic effect in different multi-drug resistant strains. The best model demonstrated notable predictive performance with an accuracy of 80% and AUROC of 0.88 on the held-out test set. When including information on antibiotic-only effects, the performance increased to an accuracy of 95% and an AUROC of 0.99. This computational approach may accelerate the identification of novel combination therapies to address the urgent challenge of antibiotic resistance.

1. Introduction

Antibiotic resistance is a rising problem and multi-drug resistant strains led to 1.27 million deaths globally in 2019 (Murray et al., 2022). The number of newly approved antibiotics with novel mechanisms of action is also decreasing, making antibiotic resistance an even greater problem in the future. One approach to combat the scarcity of new single antibiotics is to find combinations of antibiotics with other approved drugs that potentiate the antibiotic effect. Since the number of possible combinations is far greater than can be experimentally determined, the goal of our proposed project is to build a model that predicts new antibiotic-compound synergists.

To train the model we used data from Tse *et al.* (Tse et al., 2024) who developed a high-throughput droplet-based screen, DropArray, to assess the combinatorial antibiotic effect of 22 antibiotics with 30,000 compounds in six multi-drug resistant gram-negative bacteria strains. In total, they measured 1.3 million unique strain-antibiotic-compound combinations. They identified several new compounds, such as P2-56, which potentiates the effect of the antibiotic rifampin against *Acinetobacter bamanii* and *Klebsiellla pneumoniae*. Leveraging the predictive power of machine learning models, we hope to expand possible synergistic compounds beyond those tested experimentally by Tse et al. (Tse et al., 2024).

2. Methods

2.1. Model Architecture

We tested different model architectures for binary classification of antibiotic-compound synergy (1 = synergistic, 0 = not synergistic) testing: logistic regression (1), Scikitlearn RandomForestClassifier (2), XGBoost classifier (3), a gradient-boosted decision tree (GBDT) model implemented in LightGBM (LGBMClassifier) (4), and a multi-layer perceptron classifier implemented in pytorch (5). In addition to classification models, we also trained a regression model with XGBoost (6).

2.2. Compound, Antibiotic, and Strain Embeddings

The models take concatenated tabular inputs including a representation of the small molecule compounds and antibiotics, numeric assay measurements, and categorical strain information as input. The numeric features included antibiotic concentration (abx_conc) and antibiotic-only effect (Ea_med), but we also trained the models without Ea_med to focus it more on the compound characteristics. For the bacterial strains we used one-hot encoding. For the small-molecule representation of compounds and antibiotics we tested different strategies, 1024-bit Morgan Fingerprints (radius = 2) and ChemBERTa (Ahmad et al., 2022).

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Morgan fingerprints for antibiotics and screened compounds were generated using RDKit based on their canonical SMILES representations (Rogers & Hahn, 2010). The fingerprints provide efficient and consistent embedding for the chemical context, without requiring additional dimensionality reduction or embedding layers.

For ChemBERTa, we converted each SMILES string into a 199-dimensional vector, which contains representation for complex molecular information. ChemBERTa is a transformer-based deep learning model, specifically trained on SMILES representation (Ahmad et al., 2022). Built upon the RoBERTa (Robustly Optimized BERT Pretraining Approach) architecture, ChemBERTa leverages masked language modeling (MLM) during pre-training, where it learns to predict randomly masked tokens within SMILES strings, thereby capturing the underlying chemical semantics and relationships.

As the target variable we chose the bliss score, which we either used directly in the regression task or we categorized it for classification. The bliss score is a measure of compound-antibiotic synergy and was calculated as follows by Tse *et al.* (2024) (Tse et al., 2024): Bliss = Eac - (Ea + Ec - Eac), where Eac is the combined compound and antibiotic effect, Ea is the antibiotic-only effect and Ec is the compound-only effect. We classified data into into binary classes with bliss score $\geq$ 0.3 being synergistic and bliss score $\leq$ -0.3 as not synergistic, omitting data points between the thresholds. Tse *et al.* (2024) (Tse et al., 2024) similarly classified datapoints above a bliss score of 0.3 to be synergistic compounds. Our classification of datapoints with bliss score above 0.3 and below -0.3 reduced the dataset to 7292 data points, but also provided a good balance between categories as there are only few synergistic data points (2881 positive data points). We also attempted to train classifiers on all data, marking samples with bliss score $\geq$ 0.3 as synergistic and samples below as not synergistic, but did not have enough compute to get results. To look at different dataset selections, we also did another split where we randomly took 2000 values above 0.3 and randomly chose 2000 values below 0.3 and retrained the classifiers. The regression model with XGBoost was trained on all data.

2.3. Model Training

For models (1) - (4) and (6), the data was partitioned into training and test subsets using 5-fold cross-validation, implemented via KFold from scikit-learn, with shuffling enabled and a fixed random seed (random_state=42) to ensure reproducibility. For each fold, one subset was designated as the test set, while the remaining four were used for training. For the MLP model (5), the data was split into 70% training, 10% validation, and 20% test set via SciKitLearn train_test_split with random_state 42.

Models (1) - (4) were largely trained with default hyperparameters and a fixed decision threshold of 0.5 for the classification. For logistic regression (1) we trained with max_iter = 1000, for RandomForestClassifier (2) we used n_estimators = 200, and for XGBoost (3) we used eval_metric = logloss. The LGBMClassifier (4) was trained with 500 estimators, learning rate 0.05, 64 leaves, and both sbsample and colsample_bytree set to 0.8 with the objective function set to binary classification using gradient-boosted decision trees (boosting_type = "gbdt") with the area under the receiver operating characteristic curve (AUROC) as the evaluation metric.

After some manual hyperparameter optimization, the MLP classifier (5) was trained with three fully-connected layers containing 1024 nodes, with batch normalization, ReLu activation function and dropout at 0.4 between each layer. BCEWithLogitsLoss was the loss function and the model was trained with early stopping on the validation set with a tolerance of 5 and minimum improvement threshold 0.001.

3. Results

With all parameters (including Ea) and Morgan finger print embedding, the XGBoost and LightGBM decision tree qualifiers performed the best (Figure 1 a, b). Both models showed an accuracy of 95% and an AUROC (area under the receiver operating curve) of 0.99. While these non-linear decision tree models are higher than a baseline logistic regression model, the logistic regression model still showed high performance with an accuracy of 91% and AUROC of 0.97. This suggests that some of the important features have a linear component captured in this simple model.

We noticed that Ea parameter had the highest feature importance and dropping the Ea parameter reduced model performance overall (Figure 1 a, b). The Ea parameter has a negative correlation with the bliss score (Figure 2), which intuitively makes sense, since an antibiotic with low effect by itself is more like to have compounds that potentiate the antibiotic effect compared to an antibiotic that by itself is already effective. It also makes sense given the equation used to calculate bliss score (Bliss = Eac - (Ea + Ec - Eac).Since the Ea (antibiotic-only effect) parameter is not the main focus of our project, we decided to train all models without the Ea parameter as well. We hope this exclusion of the Ea parameter helps the model training to fit based on molecular parameters rather than on whether the input antibiotic has a high measured effect.

When excluding the Ea parameter, XGBoost performed slightly better than LightGBM (XGBoost accuracy 79% and AUROC of 0.87). The logistic regression model still performed with an accuracy of 68% and an AUROC of 0.72 (Figure 1 a, b). Interestingly, the MLP classifier performed

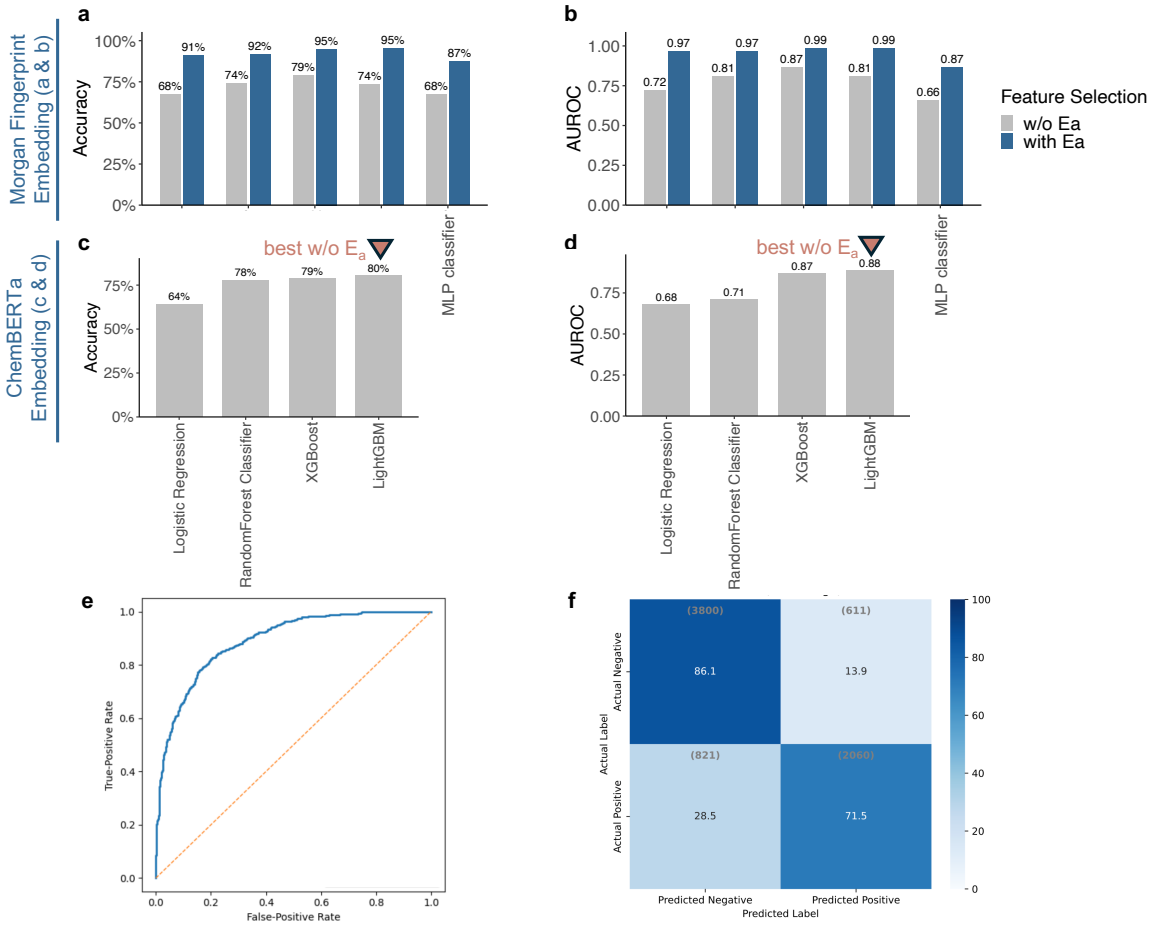

Figure 1. Classifier performance overview with LightGBM Model performing the best. (a) Accuracy of models with and without Ea parameter with Morgan Fingerprint embedding. (b) AUROC of models with and without Ea parameter with Morgan Fingerprint embedding. (c) Accuracy of models without Ea parameter with ChemBERTa embedding. (d) AUROC of models without Ea parameter with ChemBERTa embedding. (e) ROC curve for best model (LightGBM without Ea parameter with ChemBERTa embedding). (f) Confusion matrix of best model (LightGBM without Ea parameter with ChemBERTa embedding).

the worst throughout even though it is the most complex model, even after hyperparameter optimization.

To test the effect of compound embeddings, we retrained the decision tree models and the logistic regression model (models 1-4) with ChemBERTa embedding, while excluding the Ea parameter as before (Figure 1 c, d). This yielded a LightGBM model outperforming the earlier models with an accuracy of 80% and AUROC of 0.88. The ROC curve (Figure 1 e) shows that the classifier is balanced for positive and negative examples. The balanced confusion matrix (Figure 1 f) underscores the reliability of the model predictions. The confusion matrix also suggests that the model is better at avoiding false positives than false negatives at the current decision threshold of 0.5. This bias for lower false positives is advantageous in this case as predictions for compounds to be tested experimentally should not lead to many compounds without activity. A threshold shift higher than 0.5

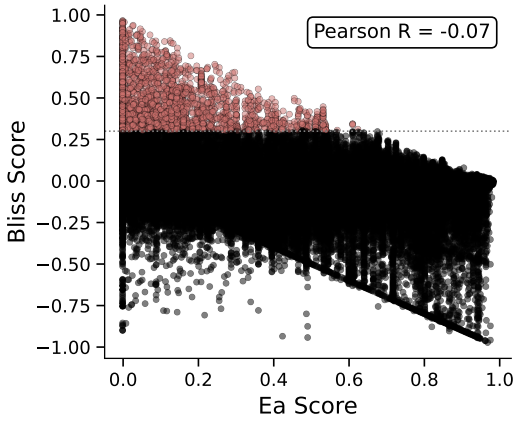

Figure 2. Ea parameter predictive of bliss score. Values with a bliss score 0.3 are marked in red as those values are classified as synergistic compounds.

could lower the false positive rate further.

Since we excluded large parts of the original dataset (7292 of 1.3 million), we also tried training the best classifier models on all data. More datapoints, however, were not feasible with our compute power. Therefore, to investigate the effect of an alternative dataset selection and distribution, we retrained the LightGBM model with Morgan Fingerprint and ChemBERTa embedding with 2000 positive and 2000 random negative dapapoints, where the negative datapoints were randomly chosen below 0.3. Therefore, this alternative training set also includes datapoints between -0.3 and 0.3 that were previously omitted. The resulting models showed slightly lower performance with an accuracy of 77% and AUROC 0.8694 for the ChemBERTa embedded data.

Besides classification, we also trained an XGBoost regressor with Morgan Fingerprint embedding (Figure 3). The regressor showed poor correlation with the actual bliss score on the held-out test set (Pearson R: 0.52, R2: 0.26, root mean squared error: 0.0584). It showed some correlation for the lower bliss scores, but did not predict anything above the synergy threshold of 0.3. Therefore, the classifier models perform much better than the regressor.

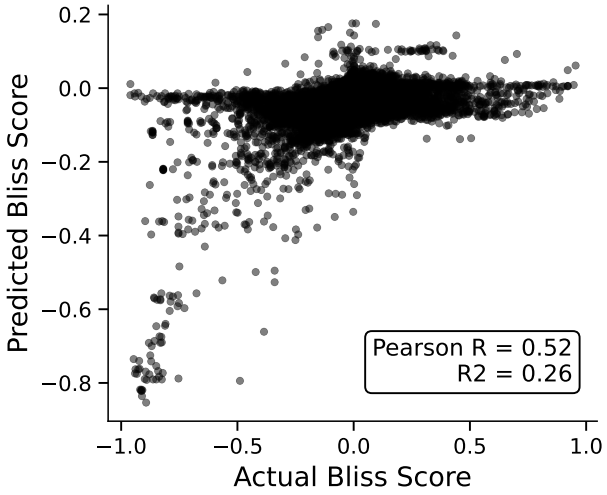

Figure 3. XGBoost regression model performance.

4. Discussion and Conclusions

We developed classifiers predicting synergistic compound-antibiotic interactions for multi-drug resistant pathogens. Our classifiers showed high accuracies, in particular the LigthGBM model with an accuracy of 80% with ChemBERTa embedding without the Ea feature.

While the classifier showed reliable performance, we envision further performance improvements through more hyperparameter optimization. Besides hyperparameter optimization, one could retrain the classifier models with more datapoints if more compute power was available. The datapoints can be increased by including more negative example when increasing the threshold from -0.3 to 0.3.

We hope our model can be used as a benchline to predict new possible compounds that could enhance antibiotic effects. To apply our model, we envision running compounds from chemical databases such as PubChem (ScienceDirect Topics) through the model in combination with all antibiotics, relevant strains and different antibiotics concentrations. One would then take the top predicted compounds. To avoid false positives and obtain a confidence of the model, one would record the probabilities of the classifier and select the best-scored compounds. New antibiotic-enhancing compounds could then be tested experimentally in the droplet-based assay described by Tse *et al.* (2024) (Tse et al., 2024).

Overall, we trained high-performing classifiers models that could suggest compound potentiating antibiotic effects.

Author contributions

Teresa Augustin - conception of the project idea, coding of logistic regression, random forest, XGBoost and MLP classifiers, writing, presentation

Gozel Dovranova - initial data processing, embedding code, LigthGBM model code, retraining of all models with ChemBERTa embedding, writing, presentation

Juhi Pandit - regression model, alternative MLP classifier, writing, presentation

Karie Shen - confusion matrix plot code, presentation

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